

Paper IV - Recent Advances (MCh Urology, Summer 2015)

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IV-1. Role of robotics in urology

Type: LAQ (25 marks)

Definition

- Robot: reprogrammable multifunctional manipulator. Current systems = **master-slave telemanipulators**; surgeon controls every movement (not autonomous).
- Urology adopted robotics earliest/widest, driven by **RARP (robot-assisted radical prostatectomy)** → now commonest RP (radical prostatectomy) approach in many high-income countries.

History

Year	Milestone
1985	PUMA 560 : CT (computed tomography)-guided brain biopsy
Late 1980s	PROBOT (Imperial College): first urological robot (automated prostate resection)
1990s	AESOP (voice-controlled laparoscope holder); ZEUS (master-slave)
2001	Lindbergh operation (Marescaux): transatlantic telesurgical cholecystectomy using ZEUS
2000	da Vinci FDA (Food and Drug Administration)-cleared; Binder & Kramer : first robotic RP
Early 2000s	Menon : VIP (Vattikuti Institute Prostatectomy) popularised RARP

da Vinci system

1. **Surgeon console**: 3D high-definition stereoscopic viewer, master controllers, foot pedals, motion scaling, tremor filtration.
2. **Patient-side cart**: 3-4 arms; **EndoWrist** instruments, 7 DOF (degrees of freedom).
3. **Vision cart**: image processing, light source, energy, assistant monitor.
 - Generations: S (2006), Si (2009; dual console, Firefly), **Xi (2014)**, **SP (single port, 2.5 cm cannula; urology 2018)**, **da Vinci 5 (2024; force feedback)**.
 - Other platforms: Hugo RAS, Versius, Senhance (haptics), **SSI Mantra (India, low-cost; telesurgery)**; cheaper competitors expected to lower costs.

Advantages vs disadvantages

Advantages	Disadvantages
3D magnified vision; 7 DOF wristed instruments	Capital ~US\$1.5–2.5 million + disposables
Tremor filtration, motion scaling, ergonomics	Limited haptics
Easier intracorporeal suturing than laparoscopy	Docking time, large footprint; surgeon remote (emergency undocking)
Less blood loss, shorter stay vs open	Learning curve, credentialing; limited public-sector access

Applications (with evidence)

1. **RARP** (index procedure)
 - **Yaxley, Lancet 2016** (RCT (randomised controlled trial), open vs robotic): urinary/sexual function **similar at 12 weeks**; robotic → less blood loss, shorter stay.
 - **Coughlin, Lancet Oncol 2018**: still **similar at 24 months**.
 - Lower transfusion, shorter stay; oncological outcomes comparable. Refinements: Retzius-sparing, NeuroSAFE, extended PLND (pelvic lymph node dissection).
2. **RAPN (robot-assisted partial nephrectomy)**: vs laparoscopic → **shorter warm ischaemia, less conversion to radical**, shorter learning curve, complex/hilar tumours; Firefly ICG (indocyanine green) selective clamping, zero ischaemia, 3D virtual models; preferred minimally invasive approach in many centres.
3. **RARC (robot-assisted radical cystectomy)**
 - **RAZOR (Parekh, Lancet 2018)**: **non-inferior 2-year PFS (progression-free survival)**; less blood loss/transfusion.
 - **iROC (Catto, JAMA 2022)**: intracorporeal diversion → **more days alive and out of hospital at 90 days**, fewer thromboembolic/wound complications.
4. **Pyeloplasty**: success >90–95%; adult, paediatric, redo PUJ (pelviureteric junction) obstruction.
5. **Ureteric reimplantation**: Boari flap/psoas hitch (adults); **RALUR** (robot-assisted laparoscopic extravesical ureteral reimplantation) for VUR (vesicoureteric reflux).
6. **Ureteric reconstruction: robotic BMG (buccal mucosa graft) ureteroplasty** for long proximal/mid strictures.
7. **R-RPLND (robotic retroperitoneal lymph node dissection)**: low-volume stage II seminoma/NSGCT (non-seminomatous germ cell tumour), selected post-chemotherapy; template preserves ejaculation.
8. Others: adrenalectomy, simple prostatectomy (>80–100 mL), sacrocolpopexy, **RAKT (robot-assisted kidney transplantation)** (fewer wound complications/lymphocele, obese recipients), IVC (inferior vena cava) thrombectomy.

Training and cost

- Didactics → **VR (virtual-reality) simulators** → dry/wet lab → supervised modular console time.
- **FRS (Fundamentals of Robotic Surgery)** curriculum; **ERUS (EAU Robotic Urology Section)** certification; dual console mentoring; institutional credentialing.
- RARP learning curve: tens of cases for proficiency; >150–250 for optimal margins/continence.
- Cost-effective only at **high volume**; cheaper platforms/reusables help (India).

Robot-specific complications

Problem	Details
Steep Trendelenburg (25-45°)	↑ IOP (intraocular pressure), ischaemic optic neuropathy , corneal abrasion, ↑ ICP (intracranial pressure), facial/laryngeal oedema, ventilatory difficulty
Positioning	Brachial plexus injury; compartment syndrome/rhabdomyolysis (prolonged lithotomy); peroneal palsy
Instrument	Insulation failure → bowel burns
Lack of haptics	Tissue tearing, suture breakage
Others	Port-site hernia, CO ₂ embolism, system failure (rare)

- Prevention: limit Trendelenburg angle/time, padding, lithotomy <2-3 h.

Recent advances

- **Single-port** extraperitoneal/transvesical RARP (same-day discharge).
- **Telesurgery** (2023-25; China, India - SSI Mantra) over 5G/fibre; latency main limit.
- **AI (artificial intelligence)** skill assessment; kinematic data predict continence; da Vinci 5 haptics.

Diagram to draw: da Vinci block diagram (console – vision cart – patient cart, EndoWrist inset) + 6-port RARP placement.

IV-2. Discuss marginal kidney donor and describe newer immunosuppressant drugs in renal transplantation

Type: LAQ (25 marks)

Introduction

- Organ shortage → **marginal donors**: higher risk of DGF (delayed graft function), primary non-function, shorter graft survival; still better survival than dialysis.

Part A: Marginal kidney donor

1. ECD (expanded criteria donor), UNOS (United Network for Organ Sharing) 2002

- Donor **≥60 years**, or **50-59 years** with **≥2** of: **hypertension**, **terminal creatinine >1.5 mg/dL**, **CVA (cerebrovascular accident) death**.
- ~70% higher relative risk of graft loss vs SCD (standard criteria donor) (relative risk >1.7).

2. KDRI (Kidney Donor Risk Index) / KDPI (Kidney Donor Profile Index)

- KDRI: 10 donor factors; KDPI = percentile. **KDPI >85%** replaced ECD (US, December 2014).

3. DCD (donation after circulatory death): Maastricht

Category	Description	Type
I	Dead on arrival	Uncontrolled
II	Unsuccessful resuscitation	Uncontrolled
III	Awaiting arrest after withdrawal of therapy	Controlled (commonest)
IV	Arrest in brain-dead donor	Either
V	Euthanasia	Controlled

- DCD: **DGF ~40-50%** but **similar long-term graft survival**.

4. Other marginal donors

- AKI (acute kidney injury)**: don't discard on terminal creatinine alone.
- HCV (hepatitis C virus)-viraemic** → **HCV-negative** + **DAAs (direct-acting antivirals)** (cure ~100%; THINKER, NEJM 2017).
- HIV (human immunodeficiency virus)-positive** → **HIV-positive** recipient (**HOPE Act 2013**).
- Living** (KDIGO (Kidney Disease: Improving Global Outcomes) 2017): older, controlled hypertension (1-2 drugs); **risk-based, projected lifetime ESKD (end-stage kidney disease) risk**.

Assessment and optimisation

- Remuzzi biopsy score** (glomeruli, tubules, interstitium, arteries; each 0-3, total 0-12):

Score	Action
0-3	Single transplant
4-6	Dual kidney transplant
≥7	Discard

- Glomerulosclerosis >20% → poorer outcome.
- HMP (hypothermic machine perfusion)**: **Moers, NEJM 2009** - ↓ **DGF (~21% vs 27%)**, ↑ 1-year graft survival; greatest benefit in ECD.
- NMP (normothermic machine perfusion)**, 37 °C: viability/repair (trials). **Cold ischaemia <12-18 h**.

- **ESP (Eurotransplant Senior Program, 1999):** donors $\geq 65 \rightarrow$ recipients ≥ 65 , short cold ischaemia over HLA (human leucocyte antigen) matching.
- Older, diabetic recipients; **informed consent**.

Outcomes

- **Ojo, JASN 2001:** ECD transplant beats dialysis. DGF: delay CNI (calcineurin inhibitor) under antibody induction.

Part B: Newer immunosuppressants

- Standard: induction (basiliximab/rATG (rabbit antithymocyte globulin)) + **tacrolimus + mycophenolate + steroids**.
- Aims: avoid CNI nephrotoxicity, $\uparrow$ adherence, desensitise, treat AMR (antibody-mediated rejection).

Induction

Drug	Mechanism	Dose / points
Basiliximab	Anti-CD25 (IL-2 (interleukin-2) receptor α), non-depleting	20 mg IV (intravenous) day 0 and day 4; low risk
rATG	Polyclonal T-cell depletion	1–1.5 mg/kg/day (total ~ 4.5 –6 mg/kg); high immunological risk, DGF; CMV (cytomegalovirus), PTLD (post-transplant lymphoproliferative disorder)
Alemtuzumab	Anti-CD52	30 mg single dose (off-label); steroid/CNI minimisation

Maintenance

Drug	Mechanism	Points
Extended-release tacrolimus (Advagraf, Envarsus XR)	Calcineurin inhibition	Once daily $\rightarrow$ adherence
mTOR (mammalian target of rapamycin) inhibitors: sirolimus, everolimus	FKBP12 $\rightarrow$ block mTOR (G1 $\rightarrow$ S)	Non-nephrotoxic, fewer skin cancers, fewer CMV/BK. TRANSFORM (Pascual, JASN 2018): everolimus + reduced CNI non-inferior . Side effects: wound healing, lymphocele , proteinuria, dyslipidaemia, pneumonitis
Belatacept	CTLA4-Ig (cytotoxic T-lymphocyte antigen 4-immunoglobulin) binds CD80/86 $\rightarrow$ blocks CD28 costimulation (signal 2)	FDA (Food and Drug Administration) 2011; 10 mg/kg (days 1, 5, weeks 2, 4, 8, 12) then 5 mg/kg every 4 weeks. BENEFIT/BENEFIT-EXT: better GFR (glomerular filtration rate) vs ciclosporin, more early rejection ; 7-year data (NEJM 2016) $\uparrow$ patient/graft survival. Contraindicated in EBV (Epstein-Barr virus)-seronegative (CNS (central nervous system) PTLD)

Desensitisation and AMR

Drug	Mechanism	Use
IVIG (intravenous immunoglobulin)	Immunomodulation	Desensitisation; AMR + plasmapheresis
Rituximab	Anti-CD20	ABO-incompatible; AMR
Bortezomib	Proteasome inhibitor (plasma cells)	AMR; BORTEJECT negative

Drug	Mechanism	Use
Eculizumab	Anti-C5	aHUS (atypical haemolytic uraemic syndrome) recurrence ; meningococcal vaccine mandatory
Imlifidase	Enzyme cleaving IgG in hours	EMA (European Medicines Agency) conditional approval 2020 , highly sensitised, positive crossmatch; 0.25 mg/kg once
Tocilizumab/clazakizumab, felzartamab	Anti-IL-6 / anti-CD38	Chronic/late AMR; investigational

- **Voclosporin**: approved for lupus nephritis (2021), **not transplantation**.
- **Tolerance**: kidney + haematopoietic stem cell (mixed chimerism).

Recent advances

- HCV-positive kidneys + DAAs; imlifidase.
- **Pig-kidney xenotransplantation** (MGH (Massachusetts General Hospital), March 2024).

Diagram to draw: three-signal T-cell activation (CD28-CD80/86 – belatacept; IL-2R-mTOR – basiliximab, mTOR inhibitors) with CNI site.

IV-3a. Etiology and management of hypospadias fistula

Type: SAQ (10 marks)

Definition and incidence

- UCF (urethrocutaneous fistula): neourethra-skin communication; **commonest complication of hypospadias repair**.
- TIP (tubularised incised plate, Snodgrass) distal **<5-10%**; proximal/staged **10-25%+**; highest in redo. Long-term follow-up (late presentation).

Aetiology

- **Distal obstruction** (meatal stenosis, stricture, narrow neourethra) → high voiding pressure: **most important correctable cause**.
- Ischaemia, infection, haematoma.
- Tension; **overlapping suture lines; no barrier layer**.
- Poor tissue: small glans (<~14 mm), shallow plate, redo, proximal hypospadias.
- Catheter problems: early removal, blockage, bladder spasm.

Prevention

- **Barrier layer:** dorsal dartos flap, **tunica vaginalis flap** (proximal/redo), spongioplasty.
- Non-overlapping suture lines, tension-free glanuloplasty, haemostasis, magnification, drainage.

Evaluation

- Site, number, size during voiding; coronal = glans dehiscence.
- **Exclude distal obstruction:** calibration, urethroscopy, uroflowmetry.
- Saline/methylene-blue injection at surgery for small/multiple tracts.

Management

- **Wait ≥6 months; correct distal obstruction same sitting.**

Fistula	Procedure
Small (<2-3 mm), single	Excise, layered closure (6-0/7-0 absorbable), dartos barrier , offset skin lines
Large/multiple/thin urethra	Redo urethroplasty (TIP, onlay, staged BMG (buccal mucosa graft))
Coronal (glans dehiscence)	Redo glanuloplasty/TIP

- Stent/catheter 5-10 days for larger/redo repairs. Simple closure succeeds **~80-90%**; recurrence → look for missed obstruction.

UCF → wait ≥6 mo → calibrate/cystoscopy
├ distal obstruction → correct same sitting
├ small single → excise + layered closure + dartos/tunica vaginalis flap
└ large/multiple/glans dehiscence → redo urethroplasty

Recent advances

- Routine barrier/tunica vaginalis flaps in proximal repairs.
- Validated outcome scores (**HOSE (Hypospadias Objective Scoring Evaluation)**, **HOPE (Hypospadias Objective Penile Evaluation)**).

Diagram to draw: cross-section – neourethra, dartos barrier, skin, offset suture lines.

IV-3b. Silodosin

Type: SAQ (10 marks)

Definition

- **Highly uroselective α 1A-adrenoceptor antagonist** for LUTS (lower urinary tract symptoms) due to BPH (benign prostatic hyperplasia).

Pharmacology

- Selectivity α 1A : α 1B ~ **162 : 1**; α 1A : α 1D ~ 55 : 1 → **minimal orthostatic hypotension**; seminal tract blockade → **ejaculatory dysfunction**.
- Bioavailability ~32%, **with a meal**; metabolised by **CYP3A4**, UGT2B7 (active glucuronide); P-gp (P-glycoprotein) substrate.

Dose and contraindications

Item	Detail
Dose	8 mg once daily (OD) with meal
Moderate renal impairment (CrCl (creatinine clearance) 30–50 mL/min)	4 mg daily
Contraindicated	CrCl <30, Child-Pugh C, strong CYP3A4 inhibitors (ketoconazole)
Caution	Other α -blockers, PDE5 (phosphodiesterase-5) inhibitors, cataract surgery

Efficacy

- **US phase III RCTs (randomised controlled trials) (Marks 2009):** ↑ IPSS (International Prostate Symptom Score) and Qmax vs placebo; onset 2–4 days.
- **Chapple, Eur Urol 2011** (vs tamsulosin 0.4 mg, 12 weeks): **non-inferior** for IPSS; **only silodosin ↓ nocturia**; more abnormal ejaculation.
- No ↓ prostate volume/AUR (acute urinary retention) risk → add 5-ARI (5 α -reductase inhibitor) for large glands.

Adverse effects

- **Anejaculation/retrograde ejaculation ~28%** (commonest, reversible; ↓ emission).
- Dizziness, orthostatic hypotension (<3%).
- **IFIS (intraoperative floppy iris syndrome):** warn before cataract surgery.

Uses

1. BPH/LUTS: first-line; elderly/cardiac patients.
2. **MET (medical expulsive therapy)**, off-label: distal stones >5 mm; SUSPEND (Lancet 2015, tamsulosin) no overall benefit; EAU (European Association of Urology) supports.
3. Stent-related symptoms; CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome).

α -blocker comparison

Drug	Selectivity	Dose	Hypotension	Ejaculatory dysfunction
Silodosin	α 1A (very high)	8 mg OD	Least	Highest (~28%)
Tamsulosin	α 1A = α 1D > α 1B	0.4 mg OD	Low	~10%
Alfuzosin	Clinically uroselective	10 mg OD	Low-moderate	Least

Drug	Selectivity	Dose	Hypotension	Ejaculatory dysfunction
Doxazosin/terazosin	Non-selective (titration)	1-8 / 1-10 mg	Higher-highest	Low

Recent advances

- Fixed-dose combinations (dutasteride, tadalafil); premature ejaculation.

Diagram to draw: $\alpha 1$ subtypes ($\alpha 1A$ prostate, $\alpha 1B$ vessels, $\alpha 1D$ bladder).

IV-3c. Sacral neuromodulation

Type: SAQ (10 marks)

See Paper III, Q3a (identical question). Recent-advance additions:

- **Rechargeable, miniaturised IPGs (implantable pulse generators):** Axonics r-SNM (sacral neuromodulation) (FDA (Food and Drug Administration) 2019), InterStim Micro.
 - **Full-body MRI (magnetic resonance imaging)-conditional** systems (1.5 T and 3 T).
 - **Implantable tibial nerve stimulators** (eCoin, Revi).
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IV-3d. Urethral stents

Type: SAQ (10 marks)

Definition

- Tubular endoprosthesis placed endoscopically in **prostatic or bulbar urethra**; minimally invasive alternative in men unfit for definitive surgery.

Classification

Type	Examples	Features
Temporary, non-epithelialising	Memokath (thermo-expandable nickel-titanium), Allium (covered nitinol)	Removable; replace over months-years
Permanent, epithelialising	UroLume	Epithelialised in 3-6 months; difficult removal; withdrawn
Biodegradable	PLGA (poly-lactic-co-glycolic acid) spirals (experimental)	Dissolve

Indications

1. BPH (benign prostatic hyperplasia) with retention/LUTS (lower urinary tract symptoms), unfit for TURP (transurethral resection of prostate).
2. **Recurrent short bulbar stricture**, unfit/unwilling for urethroplasty (avoid penile urethra).
3. **DSD (detrusor-sphincter dyssynergia) in spinal cord injury**: alternative to sphincterotomy.
4. Palliation (prostate cancer, post-radiation).

Technique

- Memokath expands ~55 °C, softens on cooling for removal.
- Stricture stent must **not cross external sphincter** (incontinence).

Complications

Complication	Comment
Migration	Commonest (temporary)
Encrustation, stones	Prostatic stents
Hyperplastic ingrowth/restenosis	UroLume; needs resection
UTI (urinary tract infection), haematuria	
Incontinence, dribbling	If crossing sphincter
Perineal pain, painful erections	Bulbar
Difficult removal	Complicates salvage urethroplasty

Results

- Prostatic: most catheter-free short term; many removed (migration, encrustation).
- UroLume: good early → high long-term **recurrent obstruction** → abandoned.
- DSD: comparable to sphincterotomy short term.

Guidelines

- **AUA (American Urological Association) 2016 (amended 2023): no stent in men considering urethroplasty.**
- **EAU (European Association of Urology):** not routine; urethroplasty remains standard.

Recent advances

- **Optilume** paclitaxel-coated balloon (FDA (Food and Drug Administration) 2021), strictures ≤ 3 cm; **ROBUST III (J Urol 2022)**: ~75% vs 27% stricture-free at 6 months vs DVIU (direct vision internal urethrotomy)/dilation (balloon, not stent).
- **iTIND** (temporary implantable nitinol device; FDA 2020): 5–7 days in prostatic urethra for BPH.

Diagram to draw: sagittal urethra – prostatic stent bladder neck to verumontanum; bulbar stent distal to external sphincter.

IV-3e. Studer Neobladder

Type: SAQ (10 marks)

Introduction

- **Orthotopic ileal low-pressure reservoir** anastomosed to urethra; **Urs Studer (Bern)**, mid-1980s.

Principles

- **Detubularisation** → disrupts peristalsis → low pressure.
- **Spherical shape (Laplace, $P = 2T/r$)** → high capacity, low pressure.
- **Isoperistaltic afferent limb** → antireflux, tension-free ureteric anastomosis.
- Voids by **Valsalva + pelvic floor relaxation**; continence via **rhabdosphincter**.

Technique

1. **54-56 cm terminal ileum**, ending **25 cm proximal to ICV (ileocaecal valve)** (bile salt/B12).
2. **Distal 40-44 cm**: opened antimesenterically, **U-shaped**, back wall sutured.
3. **Proximal 12-14 cm**: tubular **afferent limb**.
4. Ureters end-to-side (Nesbit) onto afferent limb over stents.
5. Fold into **sphere**; urethral anastomosis at **most dependent point**.
6. Capacity ~120-150 mL → **400-500 mL** over months.

Indications / contraindications

- **Indications**: radical cystectomy (MIBC (muscle-invasive bladder cancer), high-risk NMIBC (non-muscle-invasive bladder cancer)), motivated patient, **negative urethral margin**; selected women.
- **Contraindications**: **renal impairment (eGFR (estimated glomerular filtration rate) $< \sim 35-50$ mL/min or creatinine > 2 mg/dL)**, hepatic dysfunction, positive urethral margin, IBD (inflammatory bowel disease)/radiation enteritis, unable to do **CISC (clean intermittent self-catheterisation)**, incompetent sphincter.

Postoperative care

- Mucus irrigation; bicarbonate; catheter out ~10-14 days.
- Timed voiding 2 h → 3-4 h; **night alarm**.

Outcomes

- Day continence ~**90%**; night ~**70-80%**.
- CISC: $< 10-20\%$ men; **more in women** (hypercontinence up to 20-40%).

Complications

Early	Late
Urine leak, ileus, UTI (urinary tract infection), pyelonephritis	Hyperchloraemic metabolic acidosis
Mucus retention	B12 deficiency (screen from 3-5 years)
	Ureteroileal stricture ($\sim 3-10\%$) ; stones
	Incontinence/hypercontinence; pouch-vaginal fistula

Comparison

Neobladder	Features
Studer	U-pouch + afferent limb, refluxing ureters
Hautmann	~60-70 cm, W -shaped, larger initial capacity

Recent advances

- Fully intracorporeal robotic Studer; ERAS (enhanced recovery after surgery).
- Nerve/pelvic-organ-sparing cystectomy in women → ↓ hypercontinence.

Diagram to draw: marked ileal segment → spherical pouch with afferent limb.

IV-3f. Cryotherapy for carcinoma prostate

Type: SAQ (10 marks)

Definition

- Image-guided transperineal prostate freezing.

Principle

- **Joule-Thomson effect:** **argon** freezes ($\sim -185^\circ\text{C}$), **helium** thaws.
- **Thermocouples** (apex, rectal wall, sphincter); **urethral warmer**.

Generations

Generation	Features
1st	Liquid nitrogen, no imaging; high fistula/incontinence
2nd	TRUS (transrectal ultrasound), urethral warmer, thermocouples
3rd	Argon/helium, 17-gauge needles , template
4th	Computer planning; focal/hemi-gland with MRI (magnetic resonance imaging) fusion

Mechanisms of cell death

1. **Intracellular ice** (rapid freeze).
2. **Vascular injury** → **ischaemic necrosis** (main at margins).
3. Peripheral apoptosis.
4. Immunological effect. - Target **-40°C ; rapid freeze, slow thaw; two freeze-thaw cycles**.

Indications

1. **Salvage after radiation failure** (most important): biopsy-proven local recurrence, no metastases (PSMA (prostate-specific membrane antigen) PET/CT (positron emission tomography/computed tomography) or MRI), PSA (prostate-specific antigen) preferably <10 ng/mL.
2. **Primary whole-gland**: low/intermediate risk, unfit for RP (radical prostatectomy)/RT (radiotherapy), gland <40 -50 mL.
3. **Focal/hemi-gland**: MRI-visible unilateral lesion, in registry/trial.

Outcomes

- Failure = **Phoenix (nadir + 2 ng/mL)**.
- **Donnelly, Cancer 2010** (RCT (randomised controlled trial) vs EBRT (external beam radiotherapy)): **non-inferior** biochemical progression; worse sexual function.
- Salvage: 5-year biochemical disease-free ~ 50 -60%.

Complications

Complication	Primary	Salvage
ED (erectile dysfunction)	~ 80-90%	Very high
Incontinence	~ 1 -5%	~ 5 -20%
Rectourethral fistula	<0.5-1%	~ 1 -3%
Urethral sloughing	$<5\%$ (warmer)	Higher
Retention	~ 2 -5%	

Comparison

	Cryo	HIFU (high-intensity focused ultrasound)
Energy	Freezing	Focused ultrasound heat (>60 °C)
Access	Transperineal	Transrectal

Guidelines

- **EAU (European Association of Urology):** primary/focal **only in trials/registries**; salvage option in experienced centres.
- **AUA (American Urological Association)/ASTRO (American Society for Radiation Oncology) 2022:** whole-gland may be considered (low/intermediate risk).

Recent advances

- MRI-TRUS fusion focal/hemi-gland cryotherapy; PSMA PET for salvage selection.

Diagram to draw: sagittal prostate – cryoneedles, warmer, ice ball (–40 °C isotherm inside 0 °C edge).

Rapid Revision Sheet - Paper IV

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

IV-1. Role of robotics in urology (LAQ)

- PUMA 560 (1985) first surgical robot; PROBOT first urological; da Vinci FDA 2000; first RARP Binder & Kramer 2000.
- Yaxley/Coughlin: open = robotic function at 12 weeks and 24 months.
- RAZOR: RARC non-inferior 2-year PFS; iROC: intracorporeal RARC → more days alive and out of hospital.
- Steep Trendelenburg → ↑IOP/ICP, optic neuropathy; prolonged lithotomy → compartment syndrome.

IV-2. Discuss marginal kidney donor and describe newer immunosuppressant drugs in renal transplantation (LAQ)

- ECD (UNOS 2002): ≥60, or 50–59 with ≥2 of hypertension, creatinine >1.5 mg/dL, CVA death; replaced by KDPI >85% (December 2014).
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard; HMP ↓ DGF (Moers 2009); ESP ≥65 → ≥65.
- Belatacept: better GFR, contraindicated EBV-seronegative; TRANSFORM: everolimus + low CNi non-inferior.
- Imlifidase EMA 2020 for highly sensitised; voclosporin not for transplantation.

IV-3a. Etiology and management of hypospadias fistula (SAQ)

- Commonest complication; TIP distal <5–10%, proximal higher.
- Always exclude/correct distal obstruction.
- Wait ≥6 months; layered closure + barrier + non-overlapping lines.
- Coronal fistula = glans dehiscence → redo glanuloplasty.

IV-3b. Silodosin (SAQ)

- α1A : α1B ~162 : 1; 8 mg OD with meal; 4 mg if CrCl 30–50; avoid CrCl <30, Child-Pugh C, strong CYP3A4 inhibitors.
- Anejaculation ~28% (commonest); least hypotension.
- Chapple 2011: non-inferior to tamsulosin, ↓nocturia.
- Warn about IFIS before cataract surgery.

IV-3d. Urethral stents (SAQ)

- Temporary (Memokath, Allium) vs permanent epithelialising (UroLume, withdrawn).
- Indications: unfit BPH, unfit bulbar stricture, DSD.
- Problems: migration, encrustation, hyperplastic ingrowth.
- AUA: no stent if urethroplasty considered; Optilume (ROBUST III) recent advance.

IV-3e. Studer Neobladder (SAQ)

- 54–56 cm ileum, 25 cm from ICV; 40–44 cm U + 12–14 cm afferent limb.
- Detubularisation + sphere (Laplace) → low pressure, high capacity.
- Continence ~90% day, 70–80% night; women more hypercontinence/CISC.
- Acidosis, B12 deficiency; avoid if eGFR <~35–50 mL/min.

IV-3f. Cryotherapy for carcinoma prostate (SAQ)

- Argon freezes, helium thaws; –40 °C, two freeze-thaw cycles.
- Death: intracellular ice, vascular ischaemia (main), apoptosis, immune.
- Main role: salvage after RT; ED ~80–90%; fistula <1% primary, higher salvage.

- EAU: primary/focal only in trials; Phoenix = nadir + 2.